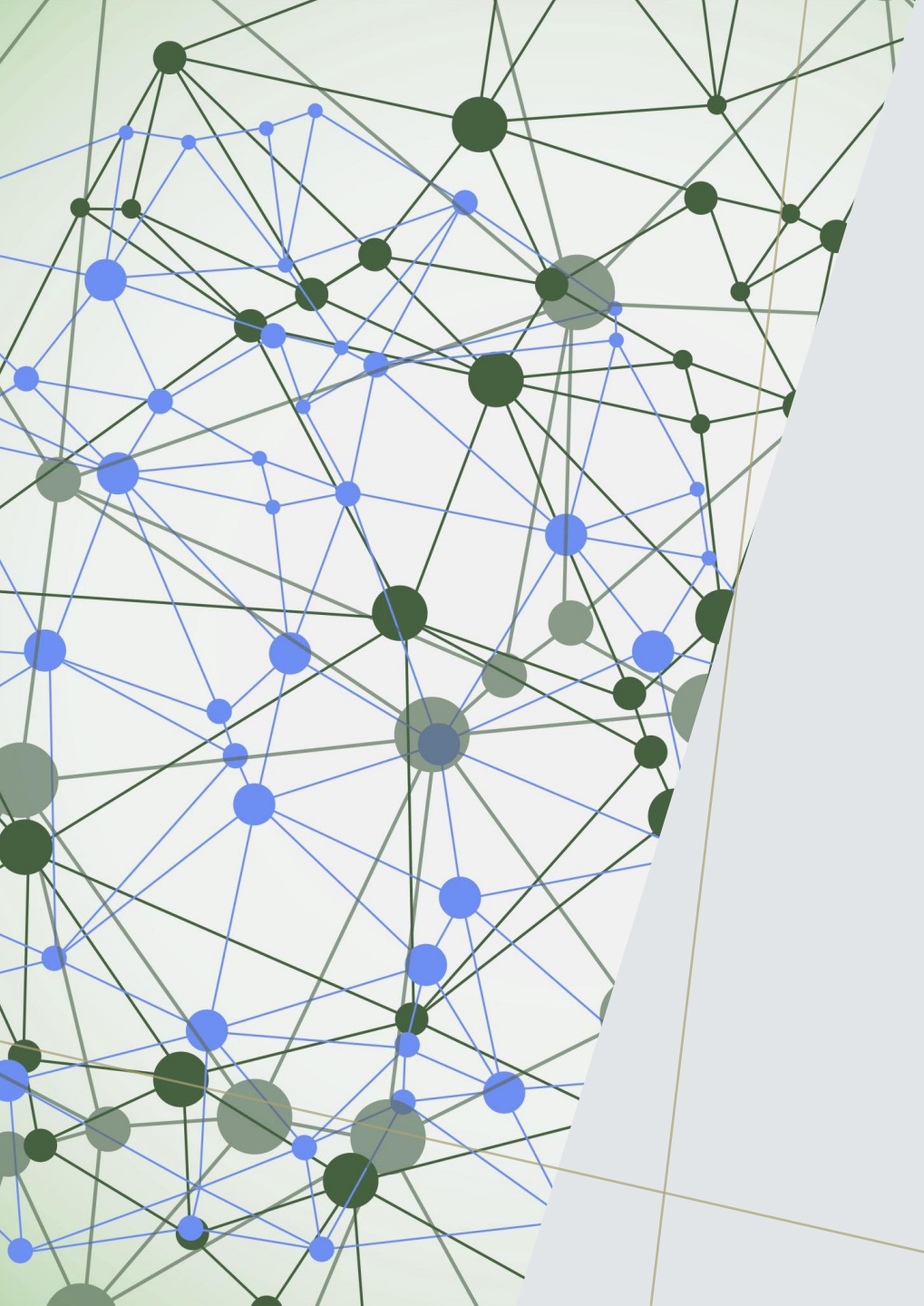


ПЛАМЕН СПАХИЕВ

ФПМИ

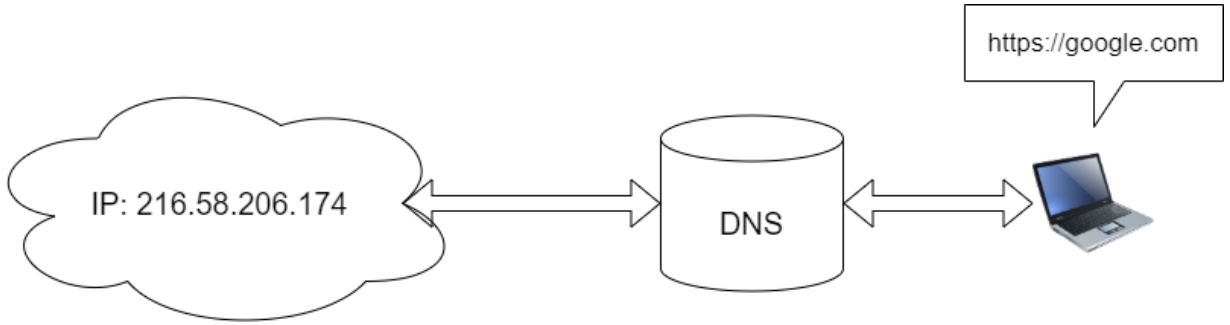
DNS

КОНЦЕПЦИЯ

11000000.10101000.00000010.00000001

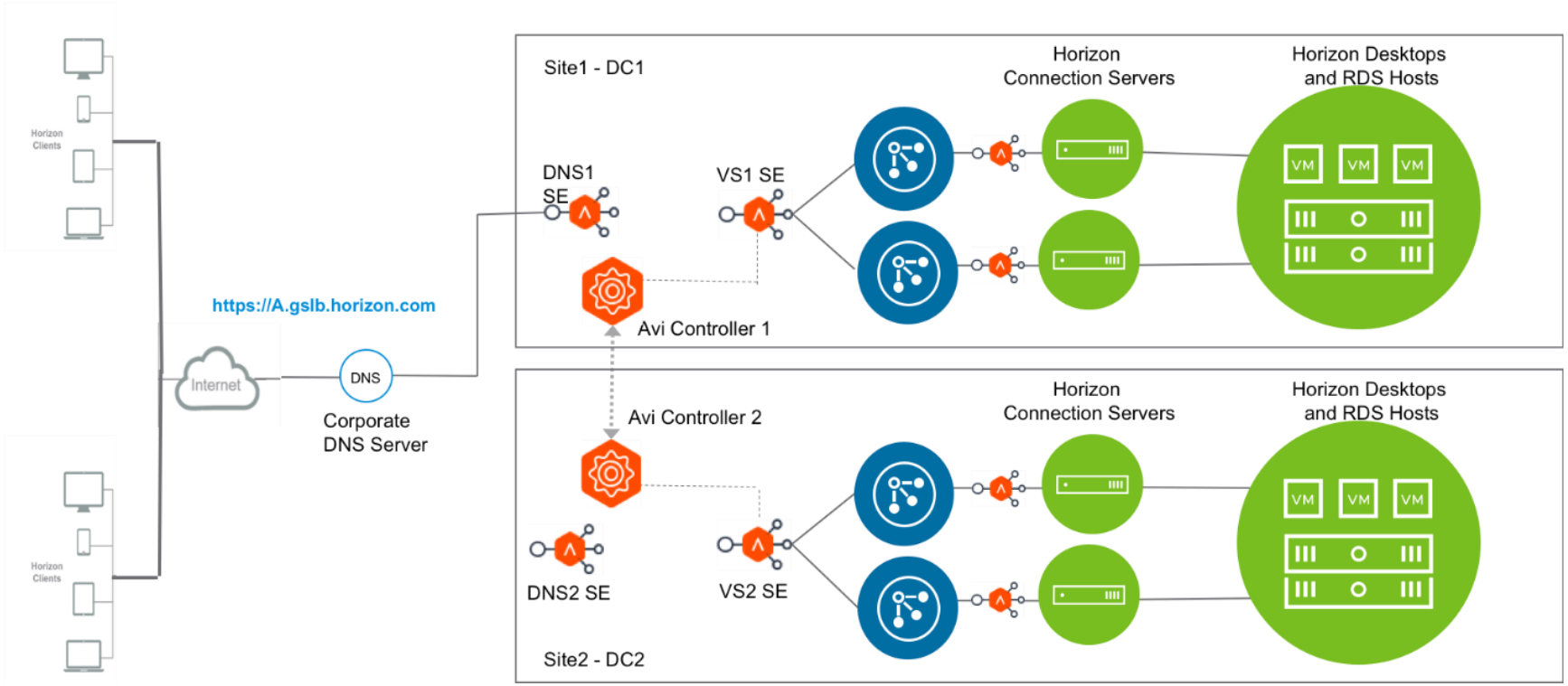
192.168.2.1

КОНЦЕПЦИЯ



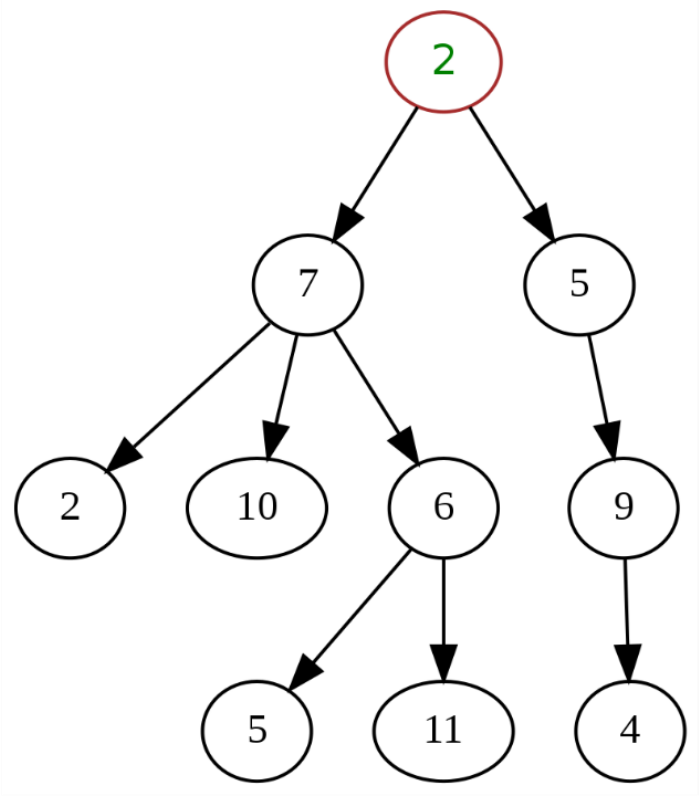
Фиг. 1. Преобразуване на имена чрез DNS системата

КОНЦЕПЦИЯ



Фиг. 2. DNS базиран DRP

СЪЩНОСТ

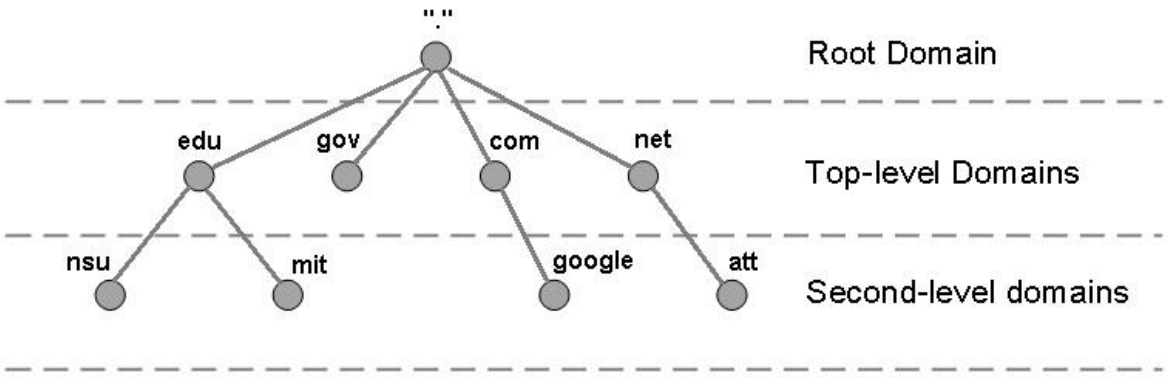


Фиг. 3. Дървовидна структура.

СЪЩНОСТ

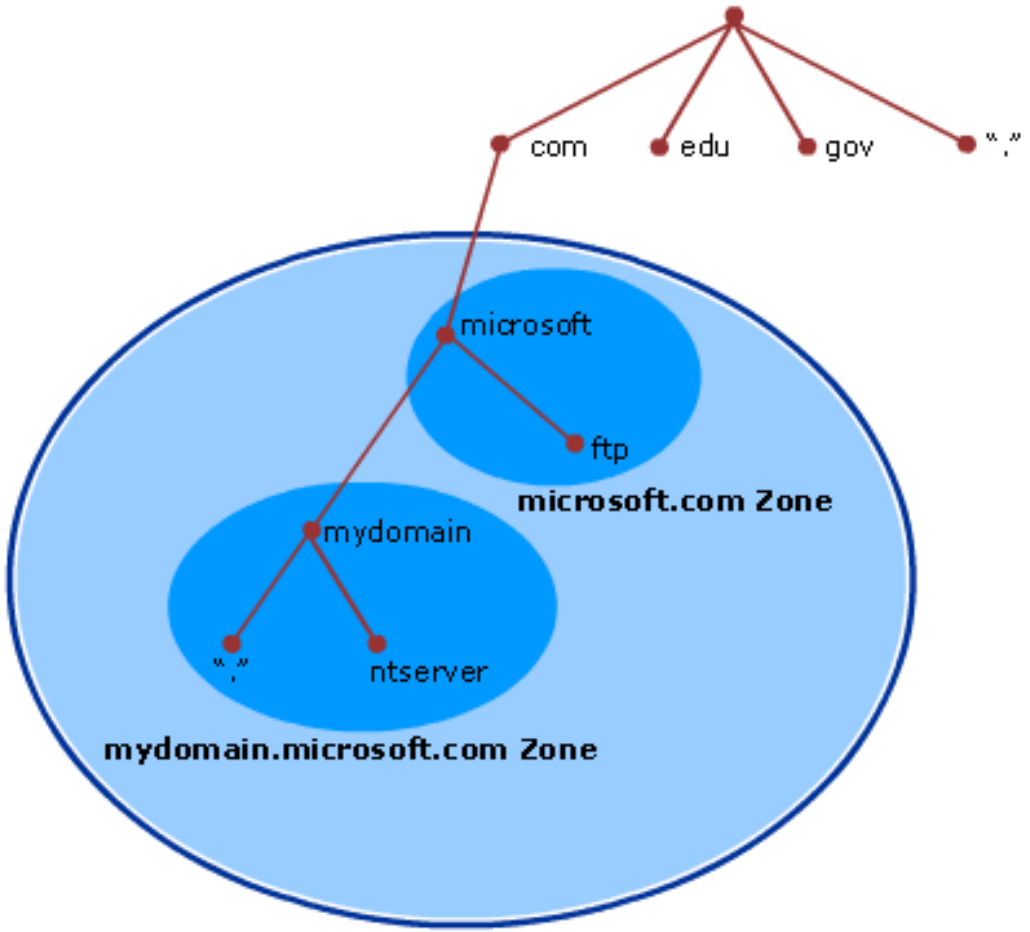
domain.com.

subdomain.domain.com



Фиг. 4. Нива в DNS йерархията

СЪЩНОСТ

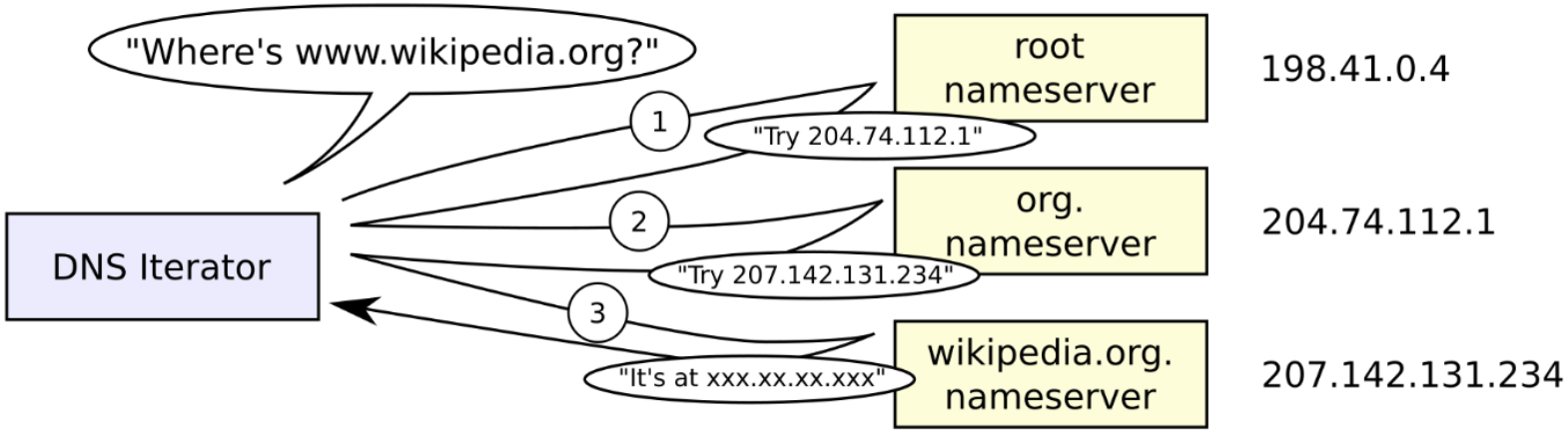


Фиг. 5. DNS зони

МЕХАНИЗЪМ НА ПРЕВРЪЩАНЕ НА ИМЕНАТА

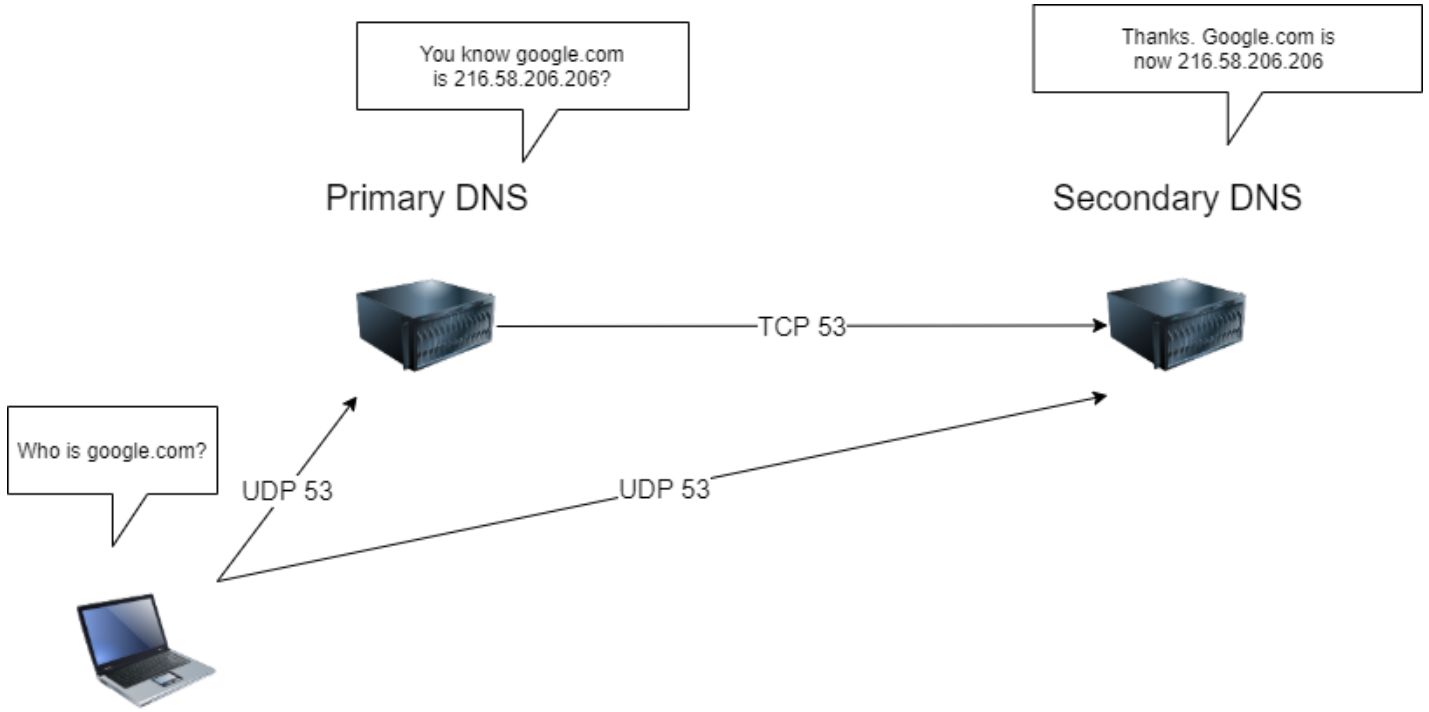
Видове резолвъри на базата на методите на запитване:

- Рекурсивни
- Нерекурсивни – authoritative отговор
- Итеративни – запитване към повече от един сървър



Фиг. 6. Итеративен подход за превръщане на имена.

МЕХАНИЗЪМ НА ПРЕВРЪЩАНЕ НА ИМЕНАТА



Фиг. 7. Транспорт при DNS

МЕХАНИЗЪМ НА ПРЕВРЪЩАНЕ НА ИМЕНАТА

Reverse lookup

142.184.250.142.in-addr.arpa

IPv4 – in-addr.arpa
IPv6 – ip6.arpa

DNS ЗАПИСИ

- **A Record** - Address record - Returns a 32-bit IPv4 address, most commonly used to map hostnames to an IP address of the host;
- **AAAA** – Address record – IPv6
- **CNAME Record** - Canonical name record - Alias of one name to another;
- **MX Record** - Mail exchange record - Maps a domain name to a list of message transfer agents for that domain.
- **TXT** – Text Record - Originally for arbitrary human-readable text in a DNS record. Since the early 1990s, however, this record more often carries machine-readable data, such as specified by RFC 1464, opportunistic encryption, Sender Policy Framework (SPF), DKIM, DMARC, DNS-SD, etc.

КОМАНДИ

```
plamen@VirtualBox:~$ host google.com
google.com has address 216.58.206.206
google.com has IPv6 address 2a00:1450:4017:802::200e
google.com mail is handled by 50 alt4.aspmx.l.google.com.
google.com mail is handled by 30 alt2.aspmx.l.google.com.
google.com mail is handled by 10 aspmx.l.google.com.
google.com mail is handled by 20 alt1.aspmx.l.google.com.
google.com mail is handled by 40 alt3.aspmx.l.google.com.
plamen@VirtualBox:~$ host google.com 8.8.8.8
Using domain server:
Name: 8.8.8.8
Address: 8.8.8.8#53
Aliases:

google.com has address 216.58.206.206
google.com has IPv6 address 2a00:1450:4017:802::200e
google.com mail is handled by 40 alt3.aspmx.l.google.com.
google.com mail is handled by 20 alt1.aspmx.l.google.com.
google.com mail is handled by 10 aspmx.l.google.com.
google.com mail is handled by 30 alt2.aspmx.l.google.com.
google.com mail is handled by 50 alt4.aspmx.l.google.com.
plamen@VirtualBox:~$
```

Фиг. 8. Прав резолв чрез командата host

КОМАНДИ

```
plamen@UirtualBox:~$ host 216.58.206.206
206.206.58.216.in-addr.arpa domain name pointer sof02s28-in-f14.1e100.net.
plamen@UirtualBox:~$ host 216.58.206.206 8.8.8.8
Using domain server:
Name: 8.8.8.8
Address: 8.8.8.8#53
Aliases:

206.206.58.216.in-addr.arpa domain name pointer sof02s28-in-f14.1e100.net.
plamen@UirtualBox:~$
```

Фиг. 9. Обратен резолв чрез командата host

КОМАНДИ

```
plamen@VirtualBox:~$ whois spahiev.com
   Domain Name: SPAHIEV.COM
   Registry Domain ID: 307262276_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.onlinenic.com
   Registrar URL: http://www.onlinenic.com
   Updated Date: 2020-12-08T14:44:02Z
   Creation Date: 2006-01-05T14:45:53Z
   Registry Expiry Date: 2023-01-05T14:45:53Z
   Registrar: OnlineNIC, Inc.
   Registrar IANA ID: 82
   Registrar Abuse Contact Email: abuse@onlinenic.com
   Registrar Abuse Contact Phone: +1 833-678-1173
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Name Server: NS3.RESELLER.BG
   Name Server: NS4.RESELLER.BG
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2021-12-13T08:07:49Z <<<
```

Фиг. 10. Резултат от whois командата

КОМАНДИ

```
plamen@UirtualBox:~$ nslookup -type=mx spahiev.com 8.8.8.8
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
spahiev.com     mail exchanger = 0 spahiev.com.

Authoritative answers can be found from:

plamen@UirtualBox:~$ nslookup -type=A spahiev.com 8.8.8.8
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   spahiev.com
Address: 79.124.16.194
```

Фиг. 11. Пример за резултат от командата nslookup

КОМАНДИ

```
plamen@VirtualBox:~$ nslookup -type=A spahiev.com NS3.RESELLER.BG
Server:         NS3.RESELLER.BG
Address:        79.124.16.194#53

Name:   spahiev.com
Address: 79.124.16.194

plamen@VirtualBox:~$
```

Фиг. 12. Пример за резултат от командата nslookup и авторитативен отговор

HOSTS FILE

```
plamen@UirtualBox:~$ cat /etc/hosts
127.0.0.1       localhost
127.0.1.1       UirtualBox

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
plamen@UirtualBox:~$
```

Фиг. 13. Съдържание на hosts файл